ASSIGNMENT NO. : 05

PRACTICAL: Jenkins Integration with GitHub for Automated Builds

Step 1: Start Jenkins

Open Jenkins in a web browser.

```
http://localhost:8080
```

Log in to the Jenkins dashboard.

Output:

```
Jenkins Dashboard opened successfully.
```

Step 2: Create a New Jenkins Project

1. Click **New Item**.
2. Enter the project name as `ai-ds-project`.
3. Select **Freestyle Project**.
4. Click **OK**.

Output:

```
New Jenkins project 'ai-ds-project' created successfully.
```

Step 3: Configure GitHub Repository

In the project configuration:

1. Go to **Source Code Management**.
2. Select **Git**.
3. Enter the GitHub repository URL. `https://github.com/<username>/ai-ds-project.git`

Output:

```
GitHub repository configured successfully.
```

Step 4: Configure Build Trigger

Go to **Build Triggers** and select:

```
GitHub hook trigger for GITScm polling
```

Output:

```
GitHub webhook trigger enabled successfully.
```

Step 5: Add Build Step

Go to **Build Steps** → **Execute Shell** and enter:

```
pip install -r requirements.txt
python train.py
```

This installs the required dependencies and runs the ML training

program. **Output:**

```
Build steps configured successfully.
```

Step 6: Create GitHub Webhook

Open the GitHub repository: **Settings** → **Webhooks** →

Add webhook Enter the Jenkins webhook URL:

```
http://<jenkins-server>:8080/github-webhook/
```

Select:

```
Content type: application/json
```

Select the **Push** event and click **Add webhook**.

Output:

```
GitHub webhook added successfully.
```

Step 7: Save Jenkins Configuration

Click **Save** and open the Jenkins project.

Output:

```
Jenkins project configuration saved successfully.
```

Step 8: Push New Code to GitHub

Make a change to the AI/DS project and push it to GitHub.

```
git add . git commit -m "Update
AI model" git push origin main
```

Output:

```
Changes pushed successfully to GitHub.
```

Step 9: Automatic Jenkins Build

After the GitHub push, the webhook automatically triggers Jenkins.
Output:

```
Started by GitHub push
Checking out code...
Installing dependencies...
Running training script...
Build completed successfully.
Finished: SUCCESS
```

Step 10: Verify Build

Open **Jenkins** → **Build History** and verify the latest build.

Output:

```
ai-ds-project #1
Build Successful
Finished: SUCCESS
```

Result

Jenkins was successfully configured with GitHub to automatically trigger a build whenever new code was pushed to the repository.

